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Electronic necklace comprising a projector medallion and method of formation thereof

Abstract

An electronic necklace comprising a projector medallion and associated methods of formation are described, and more particularly to an electronic necklace with one power source for powering a projector medallion and necklace bulbs and its associated methods of formation. In one embodiment, an apparatus may include a wire serving as a necklace strand with a projector medallion formed of a translucent material including a backside, a front side, an underside, and/or an interior. The necklace may include a power source in the form of a battery. The necklace may include a projector light and/or a medallion light within the interior of the projector medallion. The necklace may include a control chip within the interior of the projector medallion that may distribute power from the battery to the wire, the projector light, and/or the medallion light.

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Background/Summary

FIELD OF TECHNOLOGY

(1) An electronic necklace comprising a projector medallion and associated methods of formation are described, and more particularly to an electronic necklace with one power source for powering a projector medallion and necklace bulbs and its associated methods of formation.

BACKGROUND

(2) A holiday necklace may be designed to stylize a wearer with designs related to a relevant cultural event, sporting event, personal event, and/or holiday. Some holidays, including but not limited to Halloween and New Years, may have increased night-time vehicle and foot traffic on residential roads. A wearer, including but not limited to a child, may be excited and therefore may not always pay attention to his/her surroundings. The child may not know and/or ignore a rule about road safety (e.g., such as looking both ways before crossing a residential street, using designated crosswalks, and staying on sidewalks).

(3) The child's visibility on the residential street may be limited as caused by poor lighting. Insufficient and/or inadequate lighting on the residential street may make it difficult for a driver to see the child, especially during the darker evening hours of Halloween. Dimly lit areas, broken streetlights, and/or areas without streetlights altogether can create visibility challenges for both the driver and the child.

(4) Moreover, keeping with a ghoulish theme of the Halloween holiday, the child may wear a costume and/or clothing that is dark in color. This can decrease visibility, and the child may blend into a surrounding, making it harder for the driver to spot the child. This is particularly problematic at Halloween night when there may be limited ambient light.

(5) Obstructions including but not limited to parked cars, overgrown vegetation, and/or other obstacles near sidewalks can obstruct the view of the child, reducing visibility and increasing the risk of accidents with the driver. The child and the driver on the residential street may be distracted because of noises, people, costumes, and/or festivities during holiday nights. The driver may be distracted (eg, using their phone and/or engaged in other activities while behind a wheel). Therefore, the driver may not see the child.

(6) The driver may be traveling at a high speed on the residential street and may have reduced reaction time and may not have enough time to spot and avoid the child, particularly in poorly lit areas or when there are obstructions during Halloween. Adverse weather conditions during the late fall day of Halloween including but not limited to rain, fog, and/or snow can further reduce visibility for both the child and the driver, making it more challenging to see and be seen.

(7) As groups of people participate in festivities and move about a neighborhood, a front yard and/or carnival in darkened conditions, the child may be subjected to numerous risks including trips and falls and traffic accidents. Walkways and crowded areas may be difficult for the child to see and perceive because of a lack of light and/or costuming that may be visually obstructive, leading to inadvertent tripping and/or other walking accidents. Likewise, walking along and/or crossing roadways may be dangerous because of lack of light and/or dark costuming that makes the child difficult to see to drivers, leading to avoidable traffic accidents and injury.

(8) Therefore, the child may be hurt by the driver.

SUMMARY

(9) Disclosed is an electronic necklace with a projector medallion and associated methods of formation. Other features will be apparent from the accompanying drawings and from the detailed description that follows.

(10) In one aspect an apparatus includes a wire serving as a necklace strand and a projector medallion formed of a translucent material comprising a backside, a front side, an underside, and an interior. The projector medallion is connected to the necklace strand. The apparatus includes a power source in the form of a battery within the interior of the projector medallion and a projector light within the interior of the projector medallion. The apparatus includes a control chip within the interior of the projector medallion that controls power from the battery to the projector light.
(11) The apparatus includes an optical conduit within the interior of the projector medallion comprising an emblem lens and a refraction means designed to manipulate the light emanating from the projector light. The projector light is at least partially encased within the optical conduit to allow light to travel through a light canal within the optical conduit and pass through the emblem lens and the refraction means. The refraction means is at an end of the optical conduit and is adjacent to the underside of the projector medallion. The emblem lens is within the optical conduit and is positioned to receive light from the projector light.
(12) The apparatus includes an emblem integrated onto the emblem lens and an opening of the projector medallion cut into the projector medallion. The refraction means is adjacent to the opening and the optical conduit and the opening are substantially planar to one another. The opening allows light to project from the projector light to an area exterior to the refraction means without being obstructed by the translucent material of the projector medallion.
(13) The apparatus may include a medallion light within the interior of the projector medallion. The medallion light may emit light through the translucent material of the projector medallion to the area exterior to the projector medallion. The apparatus may further include a wire connector that may be designed for mechanically and/or electrically connecting the wire to the projector medallion via the control chip, which may thereby facilitate the transmission of power from the control chip to the wire while ensuring the wire's integration into the electrical system of the apparatus.
(14) The apparatus may further include a projector light connector that may provide both mechanical support and/or electrical connection for the projector light to the control chip, which may enable power transfer from the control chip to the projector light and may ensure the projector light is securely positioned within the interior of the projector medallion. The apparatus may further include a medallion light connector that may provide both mechanical support and/or electrical connection for the medallion light to the control chip, which may enable power transfer from the control chip to the medallion light and/or may ensure the medallion light is securely positioned within the interior of the projector medallion.
(15) The apparatus may further include a securing bracket that may be attached to the optical conduit and may at least partially secure the optical conduit within the interior of the projector medallion by at least partially affixing the optical conduit to the projector light connector. The apparatus may further include a proximal support, a medial support, and/or a distal support attached to a backside interior of the projector medallion which may support the optical conduit at an oblique angle to the backside interior. The proximal support may be shorter than the medial support. The medial support may be shorter than the distal support. The medial support may comprise a medial cutout to receive and/or support the optical conduit. The medial support may be closer to the underside of the projector medallion than the proximal support. The distal support may be closer to the underside of the projector medallion than the medial support and/or the proximal support. The distal support may comprise a distal cutout to receive and/or support the optical conduit.
(16) The apparatus may further include a plurality of necklace LEDs that may be attached to the wire. The plurality of necklace LEDs may receive power from the battery via the wire connector and/or the wire. The optical conduit may be polyhedral in shape. The optical conduit may be cylindrical in shape. Furthermore, when the light travels through the light canal, the light may pass through the emblem lens and/or may be at least partially obstructed by the emblem which may

create a decorative projection to the area exterior to the projector medallion after passing through the refraction means.

(17) In yet another aspect, an apparatus includes a wire serving as a necklace strand and a projector medallion formed of a translucent material comprising a backside, a front side, an underside, and an interior. The projector medallion is substantially connected to the wire. The apparatus further includes a battery compartment within the interior of the medallion and a battery serving as a power source within the battery compartment. The apparatus further includes a control chip within the interior of the projector medallion that is electrically connected to the battery and controls power to a plurality of projector lights. The plurality of projector lights are positioned within the interior.

(18) The apparatus further includes a plurality of projector light connectors that connect the plurality of projector lights to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the plurality of projector lights. The plurality of projector light connectors support the plurality of projector lights within the interior. The apparatus further includes a wire connector that connects the wire to the projector medallion via the control chip. The wire connector allows for power to move from the battery to the control chip and then to the wire. The apparatus further includes a plurality of optical conduits within the interior, each of which comprising a light canal, an emblem lens, and a refraction means. Each individual optical conduit houses at least one of the plurality projector lights.

(19) The apparatus further includes an emblem integrated onto each emblem lens. When the light travels through the light canal of the plurality of optical conduits, the light passes through the emblem lens and is at least partially obstructed by the emblem creating a decorative projection at an area exterior to the projector medallion after passing through the refraction means. The apparatus further includes a plurality of openings of the projector medallion on at least one of the front side of the projector medallion and the underside of the projector medallion. Each individual refraction means is adjacent to at least one of the openings. Each individual optical conduit of the plurality of optical conduits is substantially planar to one of the plurality of openings. Each of the plurality of openings allow light to project from the plurality of projector lights to the area exterior to the projector medallion without being obstructed by the translucent material of the projector medallion.

(20) The apparatus may include a medallion light that may be positioned within the interior and/or emits light through the translucent material of the projector medallion to the area exterior to the projector medallion. The medallion light may be electrically connected to the battery. The apparatus may further include a medallion light connector that may connect the medallion light to the control chip of the projector medallion and/or may allow for power to move from the battery to the control chip and then to the medallion light. The medallion light connector may support the medallion light within the interior.

(21) The apparatus may further include a plurality of necklace LEDs that may be attached to the wire. The plurality of necklace LEDs may be electrically connected to the battery via the necklace strand. The apparatus may further include a securing bracket attached to the plurality of optical conduits that may at least partially secure the plurality of optical conduits within the interior by at least partially affixing the plurality of optical conduits to the plurality of projector light connectors. The apparatus may further include a plurality of proximal supports, a plurality of medial supports, and/or a plurality of distal supports that may be attached to a backside interior of the projector medallion to support the plurality of optical conduits at an angle oblique to the backside interior.

(22) The plurality of proximal supports may be shorter than the plurality of medial supports and/or may be closer to the battery compartment than the plurality of medial supports. The plurality of medial supports may be shorter than the plurality of distal supports and/or may be closer to the battery compartment than the plurality of distal supports. Each of the plurality of medial supports may comprise a medial cutout to receive and/or support at least one of the plurality of optical

conduits. The plurality of distal supports may be closer to the underside of the projector medallion than the plurality of medial supports and/or the plurality of proximal supports. The plurality of medial supports may be closer to the underside of the projector medallion than the plurality of proximal supports. Each of the plurality of distal supports may comprise a distal cutout to receive and/or support at least one of the plurality of optical conduits.

(23) The apparatus may further include a plurality of necklace LEDs attached to the wire. The necklace LEDs may receive power from the battery via the control chip which may then direct power to the wire connector which may be electrically connected to the wire. The apparatus may further include a plurality of necklace bulbs that may encase the necklace LEDs, the necklace bulbs may further comprise a bulb interior and the bulb interior may further comprise an LED mount. The plurality of necklace bulbs may be formed from the translucent material. One of the plurality of necklace LEDs may be affixed within the bulb interior of one of the plurality of necklace bulbs via the LED mount. The plurality of necklace LEDs may project light through the necklace bulbs to the area exterior to the bulb.

(24) In yet another aspect, a method of manufacturing an apparatus includes forming a necklace strand from a wire. The wire is capable of transmitting an electrical current. The method includes attaching a projector medallion formed of a translucent material to the necklace strand. The projector medallion comprises a backside, a front side, an underside, and an interior and the backside comprises a backside interior that faces the interior of the projector medallion. The method further includes connecting a battery compartment to the backside interior within the interior and placing a battery serving as a power source within the battery compartment. The battery is accessible through the backside. The method further includes electrically connecting a control chip to the battery. The control chip is within the interior of the projector medallion.

(25) The method further includes securing the control chip to a chip mount within the interior. The chip mount is attached to the backside interior and is adjacent to the battery compartment. The control chip controls power from the battery to a projector light and the projector light is positioned within the interior of the medallion. The method further includes mounting the projector light to a projector light connector that connects the projector light to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the projector light. The projector light connector supports the projector light within the interior

(26) The method further includes bridging the wire connector to the control chip via a wire connector. The wire connector mounts the wire to the projector medallion at the control chip to form the necklace strand and the wire connector allows for power to move from the battery to the control chip and then to the wire. The method further includes placing an optical conduit within the interior. The optical conduit comprises a light canal, an emblem lens, and a refraction means and the optical conduit is at least one of polyhedral in shape and cylindrical in shape. The projector light is at least partially encased within the optical conduit and light from the projector light travels through the light canal of the optical conduit and passes through the emblem lens and the refraction means to an area exterior to the refraction means. The refraction means is at an end of the optical conduit and is adjacent to the underside of the projector medallion and the emblem lens is within the optical conduit at an area less close to the end of the optical conduit than the refraction means.

(27) The method further includes attaching a proximal support, a medial support, and a distal support, to the backside interior of the projector medallion to support the optical conduit at an oblique angle to the backside interior. The proximal support is shorter than the medial support and the medial support is shorter than the distal support. The medial support comprises a medial cutout to receive and support the optical conduit and the medial support is closer to the underside of the projector medallion than the proximal support. The distal support is closer to the underside of the projector medallion than the medial support and the proximal support and the distal support comprises a distal cutout to receive and support the optical conduit.

(28) The method further includes integrating an emblem onto the emblem lens. When the light

travels through the light canal, the light passes through the emblem lens and is at least partially obstructed by the emblem creating a decorative projection at the area exterior to the projector medallion after passing through the refraction means. The method further includes cutting an opening into the projector medallion on at least one of the front side of the projector medallion and the underside of the projector medallion. The refraction means is adjacent to the opening and the optical conduit and the opening are substantially planar to one another. The opening of the projector medallion is a shape that substantially corresponds to the shape of the end of optical conduit and the refraction means, and the opening allows light to project from the projector light to an area exterior to the projector medallion without being obstructed by the translucent material of the projector medallion.
(29) The method may further include attaching a securing bracket to the optical conduit that may at least partially secure the optical conduit within the interior by at least partially affixing the optical conduit to the projector light connector. The method may further include attaching a plurality of necklace LEDs to the wire. The necklace LEDs may receive power from the battery via the wire connector and/or the wire. The method may further include encasing the necklace LEDs into the plurality of necklace bulbs, the necklace bulbs may further comprise a bulb interior, the bulb interior may further comprise an LED mount. The plurality of necklace bulbs may be formed from the translucent material. One of the plurality of necklace LEDs may be affixed within the bulb interior of one of the plurality of necklace bulbs via the LED mount. The plurality of necklace LEDs may project light through the necklace bulbs to the area exterior to the bulb.
(30) The method may further include mounting a medallion light to a medallion light connector that may connect the medallion light to the control chip of the projector medallion and/or may allow for power to move from the battery to the control chip and then to the medallion light. The medallion light connector may support the medallion light within the interior.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The embodiments of this invention are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like references indicate similar elements and in which:
(2) FIG. **1** is a schematic view of an electronic necklace comprising a projector medallion, according to one embodiment.
(3) FIG. **2** is a rear view of the electronic necklace of FIG. **1** showing a backside of the projector medallion and the backside of the plurality of necklace bulbs, according to one embodiment.
(4) FIG. **3** is an underside view of the projector medallion of the electronic necklace of FIGS. **1** & **2**, according to one embodiment.
(5) FIG. **4**A is a disassembled view of the projector medallion of the electronic necklace of FIGS. **1-3** wherein the front side of the projector medallion is removed from the backside of the projector medallion and the internal components of an interior of the projector medallion are shown, according to one embodiment.
(6) FIG. **4**B is a schematic illustration of the interior of the projector medallion of the electronic necklace of FIGS. **1-4**A wherein the front side of the projector medallion is removed from the backside of the projector medallion and the internal components of the projector medallion are shown assembled, according to one embodiment.
(7) FIG. **5**A is a schematic illustration of the optical conduit of the projection projector medallion of FIGS. **1-4**B, according to one embodiment.
(8) FIG. **5**B is an alternative embodiment of FIG. **5**A wherein the emblem is inverse of the emblem shown in FIG. **5**A, according to one embodiment.

(9) FIG. **6** is a schematic illustration of the necklace bulbs of FIGS. **1-5**B wherein a bulb interior and the components of the necklace bulb are shown, according to one embodiment.
(10) FIG. **7** is an operational view of the electronic necklace of FIGS. **1-6** wherein a user is wearing the electronic necklace and the projector medallion is projecting a decorative projection downward, according to one embodiment.
(11) FIG. **8** shows an underside view of an alternative embodiment of the electronic necklace of FIGS. **1-7** wherein the projector medallion comprises two projectors, according to one embodiment.
(12) FIG. **9**A is a disassembled view of the projector medallion of the electronic necklace of FIG. **8** wherein the front side of the projector medallion is removed from the backside of the projector medallion and the internal components of an interior of the projector medallion are shown, according to one embodiment.
(13) FIG. **9**B is a schematic illustration of the interior of the projector medallion of the electronic necklace of FIGS. **8-9**A wherein the front side of the projector medallion is removed from the backside of the projector medallion and the internal components of the projector medallion are shown assembled, according to one embodiment.
(14) FIG. **10**A is an operational view of the electronic necklace of FIGS. **8-9**B wherein a user is wearing the electronic necklace and the projector medallion is projecting a first decorative projection downward, according to one embodiment.
(15) FIG. **10**B is an operational view of the electronic necklace of FIGS. **8-9**B wherein the user is wearing the electronic necklace and the projector medallion is projecting a second decorative projection downward, according to one embodiment.
(16) FIG. **11** is a process flow diagram describing a method of manufacturing the electronic necklace of FIGS. **1-10**B, according to one embodiment.
(17) Other features of the present embodiments will be apparent from the accompanying drawings and from the detailed description that follows.

DETAILED DESCRIPTION

(18) An electronic necklace comprising a projector medallion and associated methods of formation are described, and more particularly to an electronic necklace comprising one or more projectors with the projector medallion and its associated methods of use.
(19) FIG. **1** is a schematic view of an electronic necklace **100** comprising a projector medallion **102**, according to one embodiment. FIG. **1** illustrates the electronic necklace **100** comprising a projector medallion **102**, a refraction means **104**, a wire **106**, a plurality of necklace bulbs **108**A-N, a connector **110**, a front side **112**, an opening **114**, and an underside **116**.
(20) The electronic necklace **100** may be a necklace strand which may comprise the wire **106** which may further comprise electronics components for illumination. The electronic necklace **100** may be a decorative, wearable lighting accessory that combines jewelry styling with integrated illumination electronics. The electronic necklace **100** may comprise a chain-like strand with multiple necklace bulbs **108**A-N distributed along its length. The electronic necklace **100** may produce dynamic, eye-catching decorative lighting effects and/or patterns that are projected outwards, particularly from the front of the projector medallion **102**.
(21) The projector medallion **102** may be a central ornamental piece that attaches to the wire **106** of the electronic necklace **100**. The projector medallion **102** may comprise electronic components and/or projection means to create a decorative projection **702** (not shown). The projector medallion **102** may comprise metal, plastic, glass, and/or other materials and may have an exterior emblem and/or design. The projector medallion **102** may serve not only as a decorative item but may also house various components including but not limited to LEDs, a power source, and/or a projection system. The internal structure of the projector medallion **102** may be engineered to enable light projection and/or display.
(22) The refraction means **104** may be an optical element including but not limited to a lens and/or

prism. The refraction means **104** may be positioned to refract and/or direct light from within the projector medallion **102** to project the light. The refraction means **104** may work by altering the path of light emitted from one or more LED which may enable the projector medallion **102** to project patterns and/or symbols externally. The refraction means **104** may bend and/or spread the light to create a wider and/or more diffuse ornamental projection. The refraction means **104** may be made of glass, acrylic, and/or transparent plastic. The refraction means **104** may be attached to the optical conduit **404** (not shown).

(23) The wire **106** may serve as a necklace strand and/or may allow a user to wear the electronic necklace **100**. The wire **106** may be an electrical conduit to deliver power and/or signals to the various components of the electronic necklace **100**. The wire **106** may connect the projector medallion **102** to each of the necklace bulbs **108**A-N and/or carry electricity to illuminate the necklace bulbs **108**A-N. The wire **106** may also transmit control signals to coordinate blinking patterns and/or lighting effects. The wire **106** may be thin, flexible, and/or durable. The design of the wire **106** may be aesthetically pleasing which may complement the overall look of the electronic necklace **100**.

(24) The plurality of necklace bulbs **108**A-N may be individual lighting elements and/or illuminated ornaments spaced out at intervals along the length of the wire **106** of the electronic necklace **100**. Each bulb of the plurality of necklace bulbs **108**A-N may comprise a necklace LED **604** (not shown) positioned inside the bulb interior **602** (not shown). The necklace LED **604** may be illuminated by receiving power and/or signals through the wire **106**. The necklace bulbs **108**A-N may produce a decorative lighting effect that runs the length of the electronic necklace **100**. The numbers and/or spacing of the plurality of necklace bulbs **108**A-N may allow varied lighting patterns and/or effects.

(25) The connector **110** may be a clasp, hook, magnet, and/or other type of fastening mechanism which may allow the two ends of the electronic necklace **100** to be reversibly secured together and/or worn around the neck of the user **704** (not shown). The connector **110** may be connected to the wire **106** at each end to complete the circuitry running through the length of the electronic necklace **100**. The connector **110** may enable the electronic necklace **100** to form a continuous loop when worn while still being detachable for easy removal. The connector **110** may secure the electronic necklace **100** in place during use.

(26) The front side **112** may be the outward-facing frontal surface of the projector medallion **102**. The front side **112** may comprise various sub-regions including but not limited to the underside **116**. The front side **112** may comprise a decorative symbol and/or contoured shapes for producing lighting effects. The front side **112** may comprise one or more refraction means **104** and/or opening **114**. Various decorative emblems, patterns, and/or displays that light up may be exhibited on this front side **112**. The front side **112** may be an ornamental surface meant to showcase the illumination capabilities of the electronic necklace **100**.

(27) The opening **114** may be an aperture and/or port that penetrates through the underside **116** of the front side **112** of the projector medallion **102**. The opening **114** may be substantially adjacent to the refraction means **104** and/or may allow light to emanate from the projector medallion **102** to an area exterior to the projector medallion **102**.

(28) The underside **116** may be the exterior bottom surface of the projector medallion **102**. The underside **116** may face substantially downward and/or toward the ground when the electronic necklace **100** is worn by the user. The underside **116** may comprise the opening **114** and/or the refraction means **104**. The underside **116** may provide a structural base to mount internal components like LEDs, wiring connections, supports, and/or the optical conduit **404**. The underside **116** may attach to the backside **200** (not shown) to enclose the inner cavity of the projector medallion **102**.

(29) According to one embodiment, FIG. **1** shows the electronic necklace **100** which may comprise the wire **106** which may serve as a necklace strand. The wire **106** may close to form a necklace-like

loop by the connector **110**. The electronic necklace **100** may further comprise one or more necklace bulbs **108**A-N. The necklace bulbs **108**A-N may be attached to the wire **106**. The electronic necklace **100** may comprise the projector medallion **102**. The projector medallion **102** may be formed of a translucent material. The projector medallion **102** may be formed in a decorative shape. The decorative shape of the projector medallion **102** may be the same as at least one of the necklace bulbs **108**A-N. The projector medallion **102** may be connected to the wire **106** (e.g. the necklace strand) by various means including but not limited to a hook-and-loop, a snap, a clip, a screw, and/or an adhesive.

(30) The projector medallion **102** may comprise a backside **200** (not shown), the front side **112**, an underside **116**, and/or an interior **400** (not shown). The front side **112** may be attached to the backside **200** and/or the underside **116**. The front side **112** and/or the underside **116** may be commonly formed through a single mold process. The refraction means **104** may be on and/or adjacent to the front side **112** and/or the underside **116**. The front side **112** may comprise a decorative illustration. The projector medallion's **102** shape and/or orientation may be similar to and/or the same as at least one of the necklace bulbs **108**A-N. The projector medallion **102** may further comprise the opening **114** and/or the refraction means **104**. The refraction means **104** may be substantially planar with the opening **114**. The refraction means **104** and/or the opening **114** may be similarly shaped. The projector medallion **102** may further comprise a power source in the form of the battery **206** (not shown) within the interior **400** (not shown) of the projector medallion **102**.

(31) FIG. **2** is a rear view of the electronic necklace **100** of FIG. **1** showing a backside **200** of the projector medallion **102** and the backside of the plurality of necklace bulbs **108**A-N, according to one embodiment. FIG. **2** illustrates the electronic necklace **100** comprising the projector medallion **102**, the refraction means **104**, the wire **106**, the plurality of necklace bulbs **108**A-N, the connector **110**, a backside **200**, a button **202**, a battery compartment **204**, and a battery **206**. The backside **200** may be the rear surface of the projector medallion **102** that faces towards the body of the user when the electronic necklace **100** may be worn. The backside **200** may comprise functional components and/or user interfaces that may control and/or power the electronic necklace **100**.

(32) The backside **200** may provide a surface to mount items including but not limited to the button **202**, battery compartment **204**, and/or opening **114** (not shown) for wiring access to the interior components of the projector medallion **102**. The backside **200** may allow the decorative front of the electronic necklace **100** to remain unobstructed while giving the user easy access to operative parts of the electronic necklace **100**. The backside **200** may be attached to the frontside **112** by various means including but not limited to adhesives, screws, clips, hook-and-loop, and or fasteners. The backside **200** and/or the frontside **112** may be commonly formed through a single old process.

(33) The button **202** may be a user input control mounted on the backside **200** of the projector medallion **102**. The button **202** may be a switch to control the function of the projector medallion **102** and/or the various components within the interior **400** (not shown). The button **202** may be a push-button switch, a slide switch, a toggle switch, a rocker switch, a dual in-line package switch, a tactile switch, a capacitive touch sensor, a proximity sensor, a rotary switch, and/or a soft power button.

(34) By having the button **202** positioned on the backside **200**, the button **202** is easily accessible to the wearer without obstructing the front decorative view. The button **202** may serve functions including but not limited to powering the electronic necklace **100** ON/OFF and/or cycling through different illumination modes and/or patterns displayed on the front side **112**. The button **202** may send control signals to the control chip **416** (not shown). The button **202** may put the basic operation of the complex lighting effects directly at the user's fingertips on the backside **200**. The button **202** may enable adjusting the lighting without having to remove the electronic necklace **100**.

(35) The battery compartment **204** may be a housing and/or cavity located on the backside **200** of the projector medallion **102** which may comprise the battery **206** (e.g. the power source for the electronic necklace **100**). By positioning the battery compartment **204** on and/or within the

backside **200**, the battery **206** may be easily accessible to the user for swapping out depleted cells. The battery compartment **204** may be connected through wiring to power delivery components internally that distribute electricity to the LEDs, control chip **416** (not shown), and/or other powered elements. The battery compartment **204** may make the electronic necklace **100** portable by providing self-contained energy without tethering to an external power source. The location of the battery compartment **204** may prevent it from obstructing the ornamental front-side lighting effects. The battery compartment **204** may at least partially support the control chip **416** (not shown). The battery compartment **204** may be attached to the backside interior **402** (not shown).

(36) The battery **206** may be a removable and/or replaceable power source which may be housed within the battery compartment **204** of the projector medallion **102**. The battery **206** may supply the electrical energy required to illuminate the lighting components including but not limited to the projector light **406** (not shown), the medallion light **410** (not shown), and/or the necklace LEDs **604**. The battery **206** may move power to the control chip **416** and/or other electronics within interior **400** of the projector medallion **102**.

(37) The battery **206** may allow the electronic necklace **100** to be fully portable and/or untethered by providing a self-contained electricity source that does not require plugging into an external power outlet and/or supply. The battery **206** may be depleted from powering the electronic necklace **100** and/or the user may open the battery compartment **204** access point on the backside **200** to replace the existing battery **206** with a replacement battery. The battery **206** may be an alkaline battery, a lithium ion battery, a lithium polymer battery, a silver oxide battery, a zinc carbon battery, a zinc-air battery, and/or a button cell battery.

(38) According to one embodiment, FIG. **2** shows the rear portions of the various components of the electronic necklace **100**. As shown in FIG. **2**, the projector medallion **102** may comprise the backside **200**. The backside **200** may be separably attached to the frontside **112** (not shown). The backside **200** may comprise the battery compartment **204** wherein the power source (e.g. the battery **206**) may be placed. The backside **200** may further comprise the button **202** which may control the power output throughout the electronic necklace **100**. The button **202** may be placed anywhere on the projector medallion **102**. The battery **206** may be accessed through an opening and/or door on the battery compartment **204** of the backside **200** of the projector medallion **102** to swap it out with a replacement battery. The battery **206** may be integrated with the electronic components of the projector medallion **102** to provide the power source. The battery **206** may be connected to one or more of the necklace LEDs **604** of the necklace bulb **108**A-N via the wire **106**.

(39) FIG. **3** is an underside view of the projector medallion **102** of the electronic necklace **100** of FIGS. **1** & **2**, according to one embodiment. FIG. **3** illustrates the projector medallion **102** comprising the backside **200**, the refraction means **104**, the front side **112**, the underside **116**, and the opening **114**.

(40) According to one embodiment, FIG. **3** shows the frontside **112** of the projector medallion **102** connected to the backside **200**. The front side **112** may be connected to the backside **200** by various means including but not limited to glue, screws, magnets, and/or rivets. The underside **116** may be a subcomponent of the front side **112**. The electronic necklace **100** may further comprise the opening **114** of the projector medallion **102** that may be cut into the translucent material of the projector medallion **102**. The refraction means **104** may be adjacent to the opening **114**. The opening **114** may be cut into underside **116** of the frontside **112**. The opening **114** may be near and/or adjacent to the underside **116**.

(41) The optical conduit **404** (not shown) and/or the opening **114** may be substantially planar to one another. The opening **114** may allow light to project from the projector light **406** (not shown) to an area exterior to the projector medallion **102** without being obstructed by the translucent material of the projector medallion **102**. The opening **114** of the projector medallion **102** may be a shape that substantially corresponds to the shape of the end **430** (not shown) of optical conduit **404** (not shown) and/or the refraction means **104**.

(42) FIG. **4**A is a disassembled view of the projector medallion **102** of the electronic necklace **100** of FIGS. **1-3** wherein the front side **112** of the projector medallion **102** is removed from the backside **200** of the projector medallion **102** and the internal components of an interior **400** of the projector medallion **102** are shown, according to one embodiment.
(43) FIG. **4**A illustrates an interior **400** of the electronic necklace **100** comprising the refraction means **104**, the wire **106**, the backside **200**, the battery compartment **204**, the battery **206**, a backside interior **402**, an optical conduit **404**, a projector light **406**, a projector light connector **408**, a medallion light **410**, a medallion light connector **412**, a wire connector **414**, a control chip **416**, a chip mount **418**, a proximal support **422**, a medial support **424**, a distal support **426**, a securing bracket **428**, an end **430**, a medial cutout **432**, and a distal cutout **434**.
(44) The interior **400** may be the hollow inner cavity and/or space within the projector medallion **102** which may house the various electronic components that produce the decorative lighting effects. The interior **400** may provide room to mount items including but not limited the projector light **406**, the medallion light **410**, wiring connections, the control chip **416**, the battery compartment **204**, and/or structural/component supports. The interior **400** may be an enclosed environment that protects the sensitive illumination hardware which may allow all the pieces to be integrated together. The backside interior **402** may be the inner-facing portion of the backside **200**. The backside interior **402** may connect and/or support the various components of the interior **400**.
(45) The optical conduit **404** may be an elongated component that extends towards the front side **112** of the projector medallion **102**. The optical conduit **404** may be cylindrical, cuboid, and/or any other polyhedron shape. The optical conduit **404** may house the projector light **406**, the refraction means **104**, the emblem lens **504**, and/or other emblems and visual elements. The optical conduit **404** may emit light that travels through the length of the optical conduit **404**. The optical conduit **404** may use mirrors, diffusers, and/or other optics to shape the light into decorative patterns and/or images before being projected out through the refraction means **104**. The optical conduit **404** may turn point light sources into dynamic displays visible on the front **112** of the electronic necklace **100**.
(46) The projector light **406** may be one of the potentially multiple light-emitting diodes installed within the interior **400** of the projector medallion **102**. The projector light **406** may connect to circuitry to receive power and/or control signals through the projector light connector **408**. The projector light **406** may be through-hole LEDs, surface mount device (SMD) LEDs, high-power LEDs, RGB LEDs, ultraviolet LEDs, organic LEDs, chip-on-board LEDs filament LEDs, LED strips, miniature LEDs, and/or ceramic LEDs. The projector light **406** may project light through the optical conduit **404**, the emblem lens **504** (not shown), and/or the refraction means **104** and thus produce a decorative projection **702** (not shown).
(47) The projector light connector **408** may be one or more wiring terminals, sockets and/or other interfaces that the projector light **406** may attach with to integrate the projector light **406** with the control chip **416**. The projector light connector **408** may supply electrical power to the projector light **406**. The projector light connector **408** may relay control signals that may instruct the projector light **406** when to illuminate, change colors, brightness, and/or patterns based on the operating mode. The projector light connector **408** may facilitate the flexible positioning of the projector light **406** within the electronic necklace **100**.
(48) The medallion light **410** may be an additional light-emitting diode positioned within the interior **400** of the projector medallion **102**. The medallion light **410** may serve as an illumination source for producing projection effects through the front side **112**, the backside **200**, and/or the underside **116**. The medallion light **410** may create patterns, color mixing, and/or other dynamic lighting.
(49) The medallion light connector **412** may be one or more wiring terminals, sockets and/or other interfaces that the medallion light connector **412** may attach with to integrate the medallion light **412** with the control chip **416**. The medallion light connector **412** may connect the medallion light

410 to the control chip **416** of the projector medallion **102** and may allow for power to move from the battery **206** to the control chip **416** and then to the medallion light **410**. The medallion light connector **412** may facilitate the flexible positioning of the medallion light **410** within the electronic necklace **100**. The medallion light **410** may be through-hole LEDs, surface mount device (SMD) LEDs, high-power LEDs, RGB LEDs, ultraviolet LEDs, organic LEDs, chip-on-board LEDs filament LEDs, LED strips, miniature LEDs, and/or ceramic LEDs.
(50) The medallion light connector **412** may supply power to the medallion light **410** and may also relay control signals about illumination patterns, colors, and/or brightness. Having separate connectors, including but not limited to the medallion light connector **412** for the medallion light **410** and the projector light connector **408** for the projector light **406** may allow each LED to be operated independently which may facilitate the creation of more dynamic and/or vivid visual effects through the execution of distinct lighting commands for each LED in the electronic necklace **100**. The medallion light connector **412** may support the medallion light **410** within the interior **400**. The wire connector **414** may allow power and/or control signals to flow between the control chip **416** and the necklace LEDs **604** (not shown) of the necklace bulbs **108**A-N of the electronic necklace **100**.
(51) The control chip **416** may be a microcontroller, integrated circuit, and/or dedicated processor which may manage and/or coordinate all the various lighting elements and/or effects produced by the electronic components of the necklace **100**. The control chip **416** may allow for all of the lighting components to be powered by a single power source by receiving power from a power source and/or distributing the power to the various components of the necklace **100**. The control chip **416** may facilitate functions including but not limited to controlling power the projector light **406**, the medallion light **410**, and/or the necklace LEDs **604** (not shown), controlling the color and/or brightness of the projector light **406**, the medallion light **410**, and/or the necklace LEDs **604**, and executing illumination patterns and/or sequences of the projector light **406**, the medallion light **410**, and/or the necklace LEDs **604**. The control chip **416** may synchronize the different components to choreograph the complete decorative lighting experience. The control chip **416** may receive input from user controls and/or translate them into dynamic lighting shows through programmed operating modes.
(52) The chip mount **418** may be one or more of a socket, post, pin, soldered pads, platform, mount, and/or other physical connector that may emanate from the backside interior **402**. The chip mount **418** may secure the control chip **416** within the interior **400** of the projector medallion **102**. The chip mount **418** may provide mounting points to solidly affix the chip **416** in place while still allowing modular installation and/or removal if needed. The chip mount **418** may be adjacent to the battery compartment **204**. The projector medallion **102** may comprise multiple chip mounts **418** which may enable flexibility in positioning the control chip **416** optimally among the other components within the interior **400**.
(53) The proximal support **422** may be a structural brace, rail, and/or other reinforcing/structural element mounted to the backside interior **402** of the projector medallion **102**. The proximal support **422** may provide an anchoring point to firmly attach and/or position illumination components including but not limited to the optical conduit **404** and/or other electrical hardware within the interior **400**. The proximal support **422** may prevent components, including but not limited to the optical conduit **404**, from shifting out of alignment within the electronic necklace **100**. The proximal support **422** may comprise an indentation that may substantially correlate to and/or receive a portion of the optical conduit **404**.
(54) The medial support **424** may be an additional brace, rail, and/or other reinforcing/structural element which may provide extra mounting points and/or anchors for internal illumination components of the electronic necklace **100** including but not limited to the optical conduit **404** within the interior **400**. The medial support **424** may prevent components, including but not limited to the optical conduit **404**, from shifting out of alignment within the electronic necklace **100**. The

medial support **424** may comprise the medial cutout **432** that may substantially correlate to and/or receive a portion of the optical conduit **404**.
(55) The distal support **426** may be an additional brace, rail, and/or other reinforcing/structural element which may provide extra mounting points and/or anchors for internal illumination components of the electronic necklace **100** including but not limited to the optical conduit **404** within the interior **400**. The distal support **426** may prevent the optical conduit **404** from shifting out of alignment within the electronic necklace **100**. The distal support **426** may comprise the distal cutout **434** that may be substantially correlated to and/or receive a portion of the optical conduit **404**.
(56) The securing bracket **428** may be a bracket, clip, joint, and/or other fastening component that may be used to affix and/or secure the optical conduit **404** to the interior **400** of the projector medallion **102**. The securing bracket **428** may be attached to the optical conduit **404** at an area substantially opposite of the end **430**. The securing bracket **428** may hold the optical conduit **404** in place by at least partially connecting to the projector light connector **408**. The coupling between the projector light connector **408** and/or the securing bracket **428** may prevent the optical conduit **404** from coming loose over time with routine usage of the electronic necklace **100**. The securing bracket **428** may supplement the proximal support **422**, the medial support **424**, and/or the distal support **426** by adding an additional layer of reinforcement to resist vibrations and/or impacts that could displace and/or damage the various components of the electronic necklace **100**.
(57) The end **430** may be the terminating end portion and/or tip of the optical conduit **404** that extends within the interior **400** towards the underside **116** and/or front side **112** of the projector medallion **102**. The end **430** may be the illumination exit point of the optical conduit **404** where light from projector light **406** and/or other sources inside the projector medallion **102** may emerge after traveling through the length of the light canal **502** of the optical conduit **404**. The structure of the end **430** may integrate the refraction means **104** to shape the light projection before and/or as the light exits the opening **114** of the projector medallion **102**. The end **430** may transfer the illumination from the optical conduit **404** to the area exterior to the projection medallion **102**.
(58) The medial cutout **432** may be an indentation at and/or near the upper portion of the medial support **424**. The medial cutout **432** may act as a support for the optical conduit **404**. The medial cutout **432** may serve as a pathway and/or anchoring point for other components which may ensure they are correctly positioned and/or secured within the projector medallion **102.** The medial cutout **432** may match the dimensions and/or shape of the optical conduit **404** which may ensure the optical conduit **404** is secured within the interior **400** of the projector medallion **102**. Additionally, the medial cutout **432** may contribute to the overall weight reduction of the projector medallion **102**.
(59) The distal cutout **432** may be an indentation cut at and/or near the upper portion of the distal support **426**. The distal cutout **432** may act as a support for the optical conduit **404**. The distal cutout **432** of the distal support **426** may be different in shape than the medial cutout **432** of the medial support **424**. The distal cutout **432** may serve as a pathway and/or anchoring point for other components which may ensure they are correctly positioned and/or secured within the projector medallion **102**. The distal cutout **432** may match the dimensions and/or shape of the optical conduit **404** which may ensure the optical conduit **404** is secured within the interior **400** of the projector medallion **102**. Additionally, the distal cutout **432** may contribute to the overall weight distribution of the projector medallion **102**.
(60) According to one embodiment, FIG. **4**A shows the projector medallion **102** comprising various internal components. The projector medallion **102** may comprise the battery compartment **204** within the interior **400**. The projector medallion **102** may comprise a power source in the form of the battery **206** which may be stored in the battery compartment **204** within the interior **400** of the projector medallion **102**.
(61) The projector medallion **102** may comprise a control chip **416**. The control chip **416** may be

within the interior **400** of the projector medallion **102**. The control chip **416** may be adjacent to the battery compartment **204** and/or may be supported by the battery compartment **204**. The control chip may be above the battery compartment **204** (e.g. further away from the backside **200** than the battery compartment **204**). The control chip **416** may distribute power from the battery **206** to the wire **106**, the projector light **406**, and/or the medallion light **410**. The control chip **416** within the interior **400** of the projector medallion **102** may be electrically connected to a battery **206** housed within the battery compartment **204**. The control chip **216** may supply and/or control power to the projector light **406**, the medallion light **410**, and/or the plurality of necklace LEDs **604**.

(62) The projector medallion **102** may further comprise the projector light **406**. The projector light **406** may be within the interior **400** of the projector medallion **102**. The projector medallion **102** may further comprise the projector light connector **408**. The projector light connector **408** may provide both mechanical support and/or electrical connection for the projector light **406** to the control chip **416** which may enable power transfer from the control chip **416** to the projector light **406**. The projector light connector **408** may support the projector light **406** within the interior **400**. The projector light connector **408** may ensure the projector light **406** is securely positioned within the interior **400** of the projector medallion **102**. The projector light **406** connector may connect the projector light **406** to the control chip **416** of the projector medallion **102** and may allow for power to move from the battery **206** to the control chip **416** and then to the projector light **406**.

(63) The projector medallion **102** may further comprise the medallion light **410** within the interior **400** of the projector medallion **102**. The medallion light **410** may emit light through the translucent material of the projector medallion **102** to an area exterior to the projector medallion **102**. The projector medallion **102** may further comprise the medallion light connector **412** which may provide both mechanical support and/or electrical connection for the medallion light **410** to the control chip **416**. The medallion light connector **412** may enable power transfer from the control chip **416** to the medallion light **410**. The medallion light connector **412** may ensure the medallion light **410** is securely positioned within the interior **400** of the projector medallion **102**.

(64) The projector medallion **102** may further comprise the optical conduit **404**. The optical conduit **404** may be within the interior **400** of the projector medallion **102**. The optical conduit **404** may be polyhedral in shape and/or cylindrical in shape. The optical conduit **404** may comprise the emblem lens **504** (not shown) and/or the refraction means **104**. The projector light **406** may be at least partially encased within the optical conduit **404** which may allow for light from the projector light **406** to travel through the light canal **502** (not shown) within the optical conduit **404** and/or subsequently pass through the emblem lens **504** and/or the refraction means **104** to the area exterior to the refraction means **104** and/or projector medallion **102** after passing through the refraction means **104**. The refraction means **104** may be designed to manipulate the light path of the light from the projector light **406**.

(65) The refraction means **104** may be at an end **430** of the optical conduit **404**. The refraction means **104** may be adjacent to the underside **116** of the projector medallion **102**. The emblem lens **504** may be within the optical conduit **404** and/or may be positioned to receive light from the projector light **406**. The emblem lens **504** may be at an area less close to the end **430** of the optical conduit **404** than the refraction means **104**. The optical conduit **404** and/or the opening **114** may be substantially planar to one another. The opening **114** may allow light to project from the projector light **406**, through the optical conduit **404** to an area exterior to the projector medallion **102** and/or the refraction means **104** without being obstructed by the translucent material of the projector medallion **102**. The opening **114** of the projector medallion **102** may be a shape that substantially corresponds to the shape of the end **430** of optical conduit **404** and/or the refraction means **104**.

(66) The projector medallion **102** may be substantially connected to the wire **106** via the wire connector **414**. The wire connector **414** may be within the medallion interior **400**. The wire connector **414** may connect the wire **106** to the projector medallion **102** via the control chip **416**. The wire connector **414** may be designed for mechanically and/or electrically connecting the wire

106 to the projector medallion **102** via the control chip **416**. The wire connector **414** may allow for power to move from the battery **206** to control chip **416** and then to the wire **106**. When connected to the wire connector **414**, the wire **106** may form a necklace strand that may be worn by a user **704** around the user's **704** neck. The control chip **416** may facilitate the transmission of power from the control chip **416** to the wire **106** while ensuring the wire's **106** integration into the electrical system of the apparatus.

(67) The projector medallion **102** may further comprise the securing bracket **428** which may be attached to the optical conduit **404** and/or may at least partially secure the optical conduit **404** within the interior **400** of the projector medallion **102** by at least partially affixing the optical conduit **404** to the projector light **406** connector.

(68) The projector medallion **102** may further comprise the proximal support **422**, the medial support **424**, and/or the distal support **426**, all of three of which may be attached to a backside interior **402** of the projector medallion **102**. The proximal support **422**, the medial support **424**, and/or the distal support **426** may support the optical conduit **404** at an oblique angle to the backside interior **402**. The proximal support **422** may be shorter than the medial support **424**. The medial support **424** may be shorter than the distal support **426**. The distal support **426** may be closer to the underside **116** of the projector medallion **102** than the medial support **424** and/or the proximal support **422**. The medial support **424** may be closer to the underside **116** of the projector medallion **102** than the proximal support **422**.

(69) The multiple supports **422**/**424**/**426** may be attached to and/or emanate from the backside interior **402**. The multiple supports **422**/**424**/**426** may work together to properly position the optical conduit **404**. The multiple supports **422**/**424**/**426** may work together to properly position the end **430** and/or refraction means **104** such that both components align with the opening **114** and/or thus properly project the decorative projection **702**.

(70) The medial cutout **432** and/or the distal cutout **434** may be integrated into the medial support **424** and/or the distal support **426** respectively. The medial cutout **432** and/or the distal cutout **432** may position the optical conduit **404** such that the end **430** and/or the refraction means **104** are substantially planar and/or aligned with the opening **114** (not shown) of the projector medallion **100**. The medial cutout **432** and/or the distal cutout **432** may encompass different shapes and/or may not be cut into the supports **424**/**426** at all.

(71) FIG. **4**B is a schematic illustration of the interior **400** of the projector medallion **102** of the electronic necklace **100** of FIGS. **1**-**4**A, wherein the front side **112** of the projector medallion **102** is removed from the backside **200** of the projector medallion **102** and the internal components of the projector medallion **102**, are shown assembled, according to one embodiment.

(72) FIG. **4**B illustrates the interior **400** of the electronic necklace **100** comprising the refraction means **104**, the wire **106**, the backside **200**, the battery compartment **204**, the battery **206**, the backside interior **402**, the optical conduit **404**, the projector light **406**, the projector light connector **408**, the medallion light **410**, the medallion light connector **412**, the wire connector **414**, the control chip **416**, the chip mount **418**, the proximal support **422**, the medial support **424**, the distal support **426**, the securing bracket **428**, the end **430**, the medial cutout **432**, and the distal cutout **434**.

(73) According to one embodiment, FIG. **4**B shows the optical conduit **404** which may be supported by the proximal support **422**, the medial support **424**, and/or the distal support **426**. The proximal support **422** may be shorter than the medial support **424** and/or may be closer to the battery compartment **204** than the medial support **424**. The medial support **424** may be shorter than the distal support **426** and/or may be closer to the battery compartment **204** than the distal support **426**. The medial support **424** may comprise the medial cutout **432** which may receive and/or support the optical conduit **404**. The distal support **426** may be closer to the underside **116** of the projector medallion **102** than the medial support **424** and/or the proximal support **422**. The medial support **424** may be closer to the underside **116** of the projector medallion **102** than the proximal support **422**. The distal support **426** may comprise the distal cutout **434** which may receive and/or

support the optical conduit **404**.

(74) The securing bracket **428** of the optical conduit **404** may rest on the proximal support **422** and/or the medial support **424**. The end **430** the optical conduit **404** may rest closest to the distal support **426**. The medial cutout **432** and/or the distal cutout **432** may work together to support the optical conduit **404** which may provide a balanced and/or stable support structure for the optical conduit **404**. This arrangement may ensure that the optical conduit **404** does not shift and/or tilt which may maintain the desired angle and/or position for optimal light projection.

(75) The optical conduit **404** may be attached to the projector light connector **408** by guiding the securing bracket **428** to a substantial connection to the projector light connector **408**. The securing bracket **428** may comprise a notch that may receive the projector light connector **408**. By substantially connecting the to the projector light connector **408**, the securing brackets **428** may ensure that the optical conduit **404** stays substantially secure within the interior **400**.

(76) FIG. **5**A is a schematic illustration of the optical conduit **404** of the projector medallion **102** of FIGS. **1-4**B, according to one embodiment. FIG. **5** illustrates the optical conduit **404** of the interior **400** of the electronic necklace **100** comprising the refraction means **104**, a light canal **502**, an emblem lens **504**, and an emblem **506**.

(77) The light canal **502** may be a hollow cavity and/or passageway within the optical conduit **404** of the projector medallion **102** which may guide and/or funnel the light projected from the projector light **406** towards through the optical conduit **404** to (and eventually through) the emblem lens **504**. The interior surfaces of the light canal **502** may be highly reflective and/or mirrored to efficiently channel the light without losses along its length. The light canal **502** may provide a controlled path to direct the decorative lighting patterns from the internal components to the outer front side **112** of the projector medallion **102**. The light canal **502** may comprise optics including but not limited to the emblem **506**, the emblem lens **504**, and/or the refraction means **104**.

(78) The emblem lens **504** may be an optical component made of transparent and/or translucent material like glass and/or plastic that is positioned at the exit point of the light canal **502** on the front side **112** of the projector medallion **102**. The emblem lens **504** may comprise the emblem **506**. The emblem lens **504** may receive the light funneled through the light canal **502** and/or may allow it to pass through towards the refraction means **104** while potentially providing additional focusing, diffusion, and/or filtering effects. The emblem lens **504** may magnify, invert, and/or reshape the light patterns before they become the final visible emblem illumination. The emblem lens **504** may provide a customizable optical interface to transform the raw light from the canal into the electronic necklace **100**.

(79) The emblem **506** may be a decorative visual design, logo, and/or ornamental element integrated onto the emblem lens **504** that may become illuminated and/or visible as the decorative projection **702** (not shown) when the light from the optical conduit **404** travels through the light canal **502** and/or emblem lens **504** to an area outside of the medallion **102**. The emblem **506** may be etched, glued, drawn, printed, and or otherwise integrated onto the emblem lens **504** to selectively allow light to pass through the area on the emblem lens **504**. Conversely, the emblem **506** may be a shape formed on the emblem lens **504** that is etched, printed, and/or otherwise formed as a clear opening on the emblem lens **504**. In this configuration, the area on the emblem lens **504** surrounding the emblem **506** may be darker than the emblem **506** which may allow light in the form and/or shape of the emblem **506** to pass through the emblem lens **504** in its shape and/or form while the surrounding area remains substantially dark.

(80) According to one embodiment, FIG. **5** shows the optical conduit **404**. The optical conduit **404** may be within the interior **400** of the projector medallion **102**. The optical conduit **404** may comprise the emblem lens **504** and/or the refraction means **104**. The emblem lens **504** may be within the optical conduit **404** and/or may be positioned to receive light from the projector light **406**. The emblem lens **504** may be at an area less close to the end **430** of the optical conduit **404** than the refraction means **104**. The emblem lens **504** may comprise an emblem **506** that may be

integrated onto the emblem lens **504**.
(81) When the light travels through the light canal **502**, the light may pass through the emblem lens **504** and may be at least partially obstructed by the emblem **506**. The emblem **506** may create the decorative projection **702** (not shown) to the area exterior to the projector medallion **102** after passing through the refraction means **104**. The refraction means **104** may be designed to manipulate the light path of the light from the projector light **406**. The projector light **406** may be at least partially encased within the optical conduit **404** which may allow for light from the projector light **406** to travel through the light canal **502** within the optical conduit **404** and/or subsequently pass through the emblem lens **504** and/or the refraction means **104**. The refraction means **104** may be at an end **430** of the optical conduit **404** and/or may be adjacent to the underside **116** of the projector medallion **102**.
(82) FIG. **5**B is an alternative embodiment of FIG. **5**A wherein the emblem **506** is inverse of the emblem **506** shown in FIG. **5**A. FIG. **5**B illustrates the optical conduit **404** of the interior **400** of the electronic necklace **100** comprising the refraction means **104**, the light canal **502**, the emblem lens **504**, and the emblem **506**. As shown in FIG. **5**B, the emblem **506** may be oriented to allow light to pass through the emblem lens **504** in the form of a pumpkin. Converse to FIG. **5**A, the emblem in FIG. **5**B may block the light for the pumpkin's eyes, nose, mouth and/or exterior, whereas FIG. **5**A may allow the light passing through the light canal **502** to form a mouth, eyes, nose, and/or perimeter.
(83) FIG. **6** is a schematic illustration of the necklace bulbs **108**A-N of FIGS. **1**-**5**B wherein a bulb interior **602** and the components of the necklace bulbs **108**A-N are shown, according to one embodiment. FIG. **6** illustrates a bulb interior **602** of the necklace bulb **108**A-N comprising the wire **106**, a bulb interior **602**, a necklace led **604**, a bulb back **606**, a bulb face **608**, and an LED mount **610**.
(84) The bulb interior **602** may be the hollow inner cavity and/or chamber enclosed within each of the necklace bulbs **108**A-N distributed along the length of the wire **106** of the electronic necklace **100**. The bulb interior **602** may provide an insulated environment to house the necklace LED **604** and/or other lighting components for that individual the necklace bulb **108**A-N unit. The bulb interior **602** may comprise optics, reflectors, and/or diffusion materials to shape and project the necklace LED's **604** illumination outwards through the bulb face **608** and/or bulb back **606**. The bulb interior **602** may create a contained space which may separate the electronics from the outside while still enabling the light to be visible externally.
(85) The necklace LED **604** may be a compact light emitting diode positioned within the bulb interior **602** of each individual necklace bulb **108** along the electronic necklace **100**. The necklace LED **604** may be powered by receiving electricity from the wire **106**. The necklace LED **604** may produce illumination that ultimately creates the decorative lighting effect emanating from the corresponding necklace bulb **108**A-N. The necklace LED **604** may connect to the LED mount **610**. The necklace LED **604** may connect to the control chip **416** via the wire **106** and the wire connector **414** of the projector medallion **102**. The necklace LED **604** in each necklace bulb **108** may enable patterns, sequences, and/or dynamic lighting effects beyond just simple steady illumination.
(86) The bulb back **606** may be the rear outer shell and/or surface of the necklace bulb **108** that faces inwards towards the body of the user and/or neck when the electronic necklace **100** is properly worn. The bulb back **606** may enclose and/or protect the bulb interior **602** by housing the necklace LED **604** from outside interference. The bulb back **606** may be made up of a translucent and/or opaque material that may allow for light from the necklace LED **604** to illuminate to an area exterior to the necklace bulb **108**. The bulb back **606** may secure the necklace LED **604** components in place within the hollow bulb interior **602**.
(87) The bulb face **608** may be the forward-facing outer surface of the necklace bulb **108** that may be visible to outside observers when the electronic necklace **100** is properly worn by a user **704**.

The bulb face **608** may be constructed from a translucent and/or transparent plastic, acrylic and/or glass material which may allow for light to illuminate from the necklace LED **604** through the bulb face **608** to an area exterior to the necklace bulb **108**A-N. The shape, texture, and/or optical properties of the bulb face **608** may diffuse, focus, and/or otherwise alter the raw light into a more visually appealing glow and/or effect. The bulb face **608** may comprise one or more decorative emblems and/or refraction means that may create a decorative silhouette at an area exterior to the necklace bulb **108**A-N.
(88) The LED mount **610** may be the physical attachment point, socket, and/or solder pad where the necklace LED **604** becomes integrated into the bulb interior **602** of the necklace bulb **108**. The LED mount **610** may fix the necklace LED **604** in the proper orientation to project light outwards through the bulb face **608**. The LED mount **610** may further anchor the necklace LED **604** to prevent it from coming loose and/or misaligning during routine usage of the electronic necklace **100**.
(89) According to one embodiment, FIG. **6** shows the bulb interior **602** of one of the necklace bulbs **108**A-N. The electronic necklace **100** may comprise the plurality of necklace LEDs **604** which may be attached to the wire **106**. The plurality of necklace bulbs **108**A-N may be formed from a translucent material. The necklace bulbs **108**A-N may comprise the bulb interior **602**. The bulb interior **602** may further comprise an LED mount **610**. The plurality of necklace bulbs **108**A-N may encase the necklace LED(s) **604**. One of the necklace LEDs **604** may be affixed within the bulb interior **602** of one of the plurality of necklace bulbs **108**A-N via the LED mount **610**. The plurality of necklace LEDs **604** may project light through the necklace bulbs **108**A-N to an area exterior to the necklace bulb **108**A-N. The necklace LEDs **604** may receive power from the battery **206** via the control chip **416** which may then direct power to the wire connector **414** which is electrically connected to the wire **106**.
(90) FIG. **7** is an operational view of the electronic necklace **100** of FIGS. **1-6** wherein a user **704** is wearing the electronic necklace **100** and the projector medallion **102** is projecting a decorative projection **702** downward, according to one embodiment. FIG. **7** illustrates a user **704** wearing the electronic necklace **100** wherein the projector medallion **102** is producing a decorative projection **702**. FIG. **7** shows the projector medallion **102**, the wire **106**, the necklace bulbs **108**A-N, a decorative projection **702**, and a user **704**.
(91) The decorative projection **702** may be the ornamental lighting display that is produced and/or made visible on the exterior of the projector medallion **102** when the electronic necklace **100** is powered on. The decorative projection **702** may be the culminating visual effect combining the projection from the projector light **406** through the light canal **502**, emblem lens **504**, emblem **506**, and/or refraction means **104**. The decorative projection **702** may be an artistic representation that transforms the electronic necklace **100** into a wearable lighting display.
(92) The user **704** may be the individual who is wearing the electronic necklace **100** during use. The user **704** may have the necklace around his neck using the connector **110**, with the front side **112** and decorative projection **702** visible to others on the exterior surface. The backside **200** of the projector medallion **102** may comprise the button **202** and/or battery compartment **204** both of which may face inwards toward the user **704** for convenient access. The user **704** may control the lighting modes and/or effects produced in the decorative projection **702** through these interfaces on the backside **200**.
(93) According to one embodiment, FIG. **7** shows the user **704** wearing the electronic necklace **100**. When wearing the electronic necklace **100**, the electronic necklace **100** may be turned on, which may allow for a decorative protection **702** to emanate from the projection medallion **102**. The decorative projection **702** may form on the ground area near the user **704** and/or may move as the user **704** and/or the electronic necklace **100** may move. The necklace bulbs **108**A-N may also be emanating light from the necklace LED(s) **604**.
(94) FIG. **8** shows an underside view of an alternative embodiment of the electronic necklace **100**

of FIGS. **1-7** wherein the projector medallion **802** comprises two projectors, according to one embodiment. FIG. **8** illustrates the projector medallion **802** comprising a refraction means **804**A, a refraction means **804**B, an opening **806**A, an opening **806**B, a backside **810**, a front side **808**, and an underside **812**.

(95) According to one embodiment, FIG. **8** shows the front side **808** of the projector medallion **802** which may be connected to the backside **810**. The front side **808** may be connected to the backside **810** by various means including but not limited to glue, screws, magnets, and/or rivets. The underside **812** may be a subcomponent of the front side **808**. The projector medallion **802** may further comprise the opening **806**A and/or the opening **806**B both of which may be cut into the translucent material. The refraction means **804**A may be adjacent to the opening **806**A. The refraction means **804**B may be adjacent to the opening **806**B. The opening **806**A and/or the opening **806**B may be cut into the front side **808** at and/or near the underside **812**. Each individual optical conduit of the plurality of optical conduits **904**A/**904**B (not shown) may be substantially planar to one of the plurality of openings **806**A/**806**B. Each individual optical conduit **904**A/**904**B (not shown) of the plurality of optical conduits **904**A/**904**B (not shown) may house at least one of the plurality projector lights **906**A/**906**B (not shown). The openings **806**A and/or **806**B may allow light to project from the projector light **906**A and **906**B (not shown) to an area exterior to the projector medallion **802** which may occur without being obstructed by the translucent material. The openings **806**A and/or **806**B may be shaped to substantially correspond to the shape of the end (not shown) of the optical conduit **904**A and **904**B (not shown) and/or the refraction means **804**A and/or **804**A.

(96) FIG. **9**A is a disassembled view of the projector medallion **802** of the electronic necklace **100** of FIG. **8** wherein the front side **808** of the projector medallion **802** is removed from the backside **810** of the projector medallion **802** and the internal components of an interior **900** of the projector medallion **802** are shown, according to one embodiment.

(97) FIG. **9**A illustrates an interior **900** of the electronic necklace **100** comprising the, the wire **106**, refraction means **804**A, the refraction means **804**B, the backside **810**, a backside interior **902**, an optical conduit **904**A, an optical conduit **904**B, a projector light **906**A, a projector light **906**B, a first LED connector **908**A, a first LED connector **908**B, a medallion LED **910**, a second LED connector **912**, a wire connector **914**, a control chip **916**, a chip mount **918**, a proximal support **920**A, a proximal support **920**B, a medial support **922**A, a medial support **922**B, a distal support **924**A, a distal support **924**B, a securing bracket **926**A, a securing bracket **926**B, an end **928**A, an end **928**B, a medial cutout **930**A, a medial cutout **930**B, a distal cutout **932**A, a distal cutout **932**B, a battery compartment **934**, and a battery **936**.

(98) According to one embodiment, FIG. **9**A shows the projector medallion **802** which may comprise the battery compartment **934** within the interior **900**. A power source in the form of the battery **936** may be stored in the battery compartment **934** within the interior **900**. The control chip **916** may be within the interior **900** and/or adjacent to the battery compartment **934**. The control chip **916** may be supported by the battery compartment **934** and/or positioned above the battery compartment **934** (e.g. further away from the backside **810**). The control chip **916** may distribute power from the battery **936** to the wire **106**, the medallion LED **910**, the projector lights **906**A and **906**B and/or other components. The medallion LED **910** may be within the interior **900** and may emit light through the translucent material of the projector medallion **802** to the exterior. The second LED connector **912** may provide mechanical support and/or electrical connection between the medallion LED **910** and the control chip **916**.

(99) For the first decorative projection **1006** (not shown), the projector medallion **802** may comprise the projector light **906**A which may be connected to the control chip **916** via the first LED connector **908**A. The projector medallion **802** may also include an optical conduit **904**A which may comprise the refraction means **804**A at the end **928**A, each of which may be aligned with an opening **806**A. The optical conduit **904**A may be supported by the proximal support **920**A, medial support **922**A, and/or the distal support **924**A at an oblique angle to the backside interior

902. The securing bracket **926**A may secure the optical conduit **904**A to the second LED connector **912**.
(100) For the second decorative projection **1008** (not shown), the projector medallion **802** may comprise the projector light **906**B which may be connected to the control chip **916** via the first LED connector **908**B. The projector medallion **802** may also include the optical conduit **904**B which may comprise the refraction means **804**B at the end **928**B, each of which may be aligned with the opening **806**B. The optical conduit **904**B may be supported by the proximal support **922**B, medial support **922**B, and/or distal support **924**B at an oblique angle to the backside interior **902**. The securing bracket **926**B may secure the optical conduit **904**B to the first LED connector **908**B.
(101) The optical conduits **904**A and **904**B may be a channel and/or waveguide that guides and/or directs the light emitted by the projector lights **906**A/**906**B towards the refraction means **804**A/**804**B. The optical conduits **904**A/**904**B may comprise a light canal that may further shape and/or collimates the light before it enters the refraction means **804**A/**804**B. The refraction means **804**A/**804**B may be attached to the optical conduits **904**A/**904**B which may act as the final optical element(s) to refract and/or project the light in the desired pattern and/or shape away from the projector medallion **802**. Additionally, an emblem lens (not shown) and/or an emblem (not shown) may be integrated within the optical conduits **904**A/**904**B and/or the refraction means **804**A/**804**B. The emblem lens(es) may be a secondary lens element that further refracts or focuses the light as it passes the optical conduits **904**A/**904**B. The emblem may be an etched, engraved, or textured surface that imparts the specific decorative pattern or design onto the projected light.
(102) The wire connector **914** may connect the wire **106** to the control chip **916** which may allow power to move from the battery **936** to the control chip **916** to the wire **106**. The control chip **916** may further move power to the medallion LED **910**, and/or projector lights **906**A and **906**B. The medial cutouts **930**A and **930**B and/or distal cutouts **932**A and **932**B may be integrated into their respective and/or corresponding supports to properly position the optical conduits **904**A and **904**B which may align the refraction means **804**A and **804**B with the openings **806**A and **806**B for projecting decorative patterns.
(103) FIG. **9**B is a schematic illustration of the interior of the projector medallion **802** of the electronic necklace **100** of FIGS. **8-9**A wherein the front side **808** of the projector medallion **802** is removed from the backside **810** of the projector medallion **802** and the internal components of the projector medallion **802** are shown assembled, according to one embodiment.
(104) FIG. **9**B illustrates an interior **900** of the electronic necklace **100** comprising the wire **106**, the refraction means **804**A, the refraction means **804**B, the backside **810**, the backside interior **902**, the optical conduit **904**A, the optical conduit **904**B, the projector light **906**A, the projector light **906**B, the first LED connector **908**, the medallion LED **910**, the second LED connector **912**, the wire connector **914**, the control chip **916**, the chip mount **918**, the first LED connector **908**B, the proximal support **920**A, the proximal support **920**B, the medial support **922**A, the medial support **922**B, the distal support **924**A, the distal support **924**B, the securing bracket **926**A, the securing bracket **926**B, the end **928**A, the end **928**B, the medial cutout **930**A, the medial cutout **930**B, the distal cutout **932**A, the distal cutout **932**B, the battery compartment **934**, and the battery **936**.
(105) According to one embodiment, FIG. **9**B shows the optical conduits **904**A/**904**B which may be supported by the proximal supports **920**A/**920**B, the medial supports **922**A/**922**A, and/or the distal supports **924**A/**924**B respectively. The proximal supports **920**A/**920**B may be shorter than the medial supports **922**A/**922**B and may be closer to the battery compartment **934**. The medial supports **922**A/**922**B may be shorter than the distal support **924**A/**924**B and may be closer to the battery compartment **934** than the distal supports **924**A/**924**B. The medial supports **922**A/**922**B may comprise the medial cutouts **930**A/**930**B. The medial cutouts **930**A/**930**B may receive and/or support the optical conduits **904**A/**904**B. The distal supports **924**A/**924**B may be closer to the underside **812** of the projector medallion **802** than the medial supports **922**A/**922**B and the proximal supports **920**A/**920**B. The medial supports **922**A/**922**B may be closer to the underside **812**

than the proximal supports **920**A/**920**B. The securing brackets **926**A/**926**B of the optical conduits **904**A/**904**B may rest on the proximal supports **920**A/**920**B and/or the medial supports **922**A/**922**B. The ends **928**A/**928**B of the optical conduits **904**A/**904**B may rest closest to the distal supports **924**A/**924**B. The medial cutouts **930**A/**930**B and/or the distal cutouts **932**A/**932**B may work together to support the optical conduits **904**A/**904**B, which may provide a balanced and stable support structure.

(106) As shown in FIG. **9**B, the optical conduit **904**A may be attached to the first LED connector **908**A by guiding the securing bracket **926**A to a substantial connection to the first LED connector **908**A. Similarly, the optical conduit **904**B may be attached to the first LED connector **908**B by guiding the securing bracket **926**B to a substantial connection to the first LED connector **908**B. The securing brackets **926**A/**926**B may comprise notches that may receive the respective first LED connectors **908**A/**908**B. By substantially connecting to the LED connectors **908**A/**908**B, the securing brackets **926**A/**926**B may ensure that the optical conduits **904**A/**904**B stay substantially secure within the backside interior **902**.

(107) The proximal support **920**A may be placed at one edge of the battery compartment **934** and/or the proximal support **920**B may be placed at the other edge of the battery compartment **204**. Both proximal supports **920**A and **920**B may be placed at a sufficient distance from each other. This overall arrangement with the supports at different levels (proximal **920**A/**920**B, medial **922**A/**922**B, distal **924**A/**924**B) and/or cutouts (medial **930**A/**930**B cutouts and distal **932**A/**932**B cutouts) may allow the optical conduits **904**A/**904**B to be securely held in the desired angle and/or position for optimal light projection from the projector lights **906**A/**906**B through the optical conduits **904**A/**904**B to the area outside of the projector medallion **802**.

(108) FIG. **10**A is an operational view of the electronic necklace **100** of FIGS. **8-9**B wherein a user **1004** is wearing the electronic necklace **100** and the projector medallion **802** is projecting a first decorative projection **1006** downward, according to one embodiment. FIG. **10**A illustrates a user **1004** wearing the electronic necklace **100** wherein the projector medallion **802** may produce the first decorative projection **1006**. FIG. **10**A shows the projector medallion **802**, the wire **106**, the necklace bulbs **1002**A-N, a first decorative projection **1006**, and a user **1004**.

(109) According to one embodiment, FIG. **10**A shows the user **1004** wearing the electronic necklace **100**. When wearing the electronic necklace **100**, the electronic necklace **100** may be turned on, which may allow for a first decorative projection **1006** to emanate from the projector medallion **802** of the refraction means **804**A. The first decorative projection **1006** may form on the ground area near the user **1004** and may move as the user **1004** and/or the electronic necklace **100** moves. The necklace bulbs **1002**A-N may also be emanating light from the necklace LEDs.

(110) FIG. **10**B is an operational view of the electronic necklace of FIGS. **8-9**B wherein the user **1004** is wearing the electronic necklace **100** and the projector medallion **802** is projecting a second decorative projection **1008** downward, according to one embodiment. FIG. **10**B illustrates the user **1004** wearing the electronic necklace **100** wherein the projector medallion **802** may produce the second decorative projection **1008**. FIG. **10**B shows the projector medallion **802**, the wire **106**, the necklace bulbs **1002**A-N, a second decorative projection **1008**, and the user **1004**.

(111) According to one embodiment, FIG. **10**A shows the user **1004** wearing the electronic necklace **100**. When wearing the electronic necklace **100**, the electronic necklace **100** may be turned on, which may allow for a second decorative projection **1008** to emanate from the projector medallion **802** of the refraction means **804**B. The second decorative projection **1008** may form on the ground area near the user **1004** and may move as the user **1004** and/or the electronic necklace **100** moves. The necklace bulbs **1002**A-N may also be emanating light from the necklace LEDs.

(112) As shown in FIG. **10**A and FIG. **10**B, In the FIG. **10**A, the first decorative projection **1006** may be visible, indicating that the refraction means **804**A may be actively projecting light from the projector light **906**A and/or creating a visual pattern and/or image on a surface and/or area in front of the user **1004**. Conversely, in FIG. **10**B, the second decorative projection **1008** may be visible,

suggesting that the refraction means **804**B may be actively projecting light from the projector light **906**B, while the refraction means **804**A may be inactive and/or turned off. The decorative projections **1006** and **1008** may be directed downward, potentially projecting onto a surface and/or area below the user's **1004** line of sight, including but not limited to tabletop and/or the ground. This downward projection direction may be intentional which may allow the user **1004** to easily view and appreciate the decorative projections **1006** and **1008** without having to look up and/or strain their neck.

(113) The ability to switch between the first decorative projection **1006** and the second decorative projection **1008** may be controlled by the control chip **916** within the projector medallion **802**. The control chip **916** may be programmed to alternate between activating projector lights **906**A/**906**B, or it may be user-controlled, allowing the wearer to switch between the decorative projections **1006**/**1008** as desired. The necklace bulbs **1002**A-N may also contribute to the overall decorative effect, either by illuminating and/or by providing a decorative shape or design along the necklace strand, complementing the decorative projections **1006** and/or **1008** from the projector medallion **802**.

(114) FIG. **11** is a process flow diagram describing a method of manufacturing the electronic necklace of FIGS. **1**A-**10**B, according to one embodiment. In operation **1102**, a necklace strand may be formed from a wire. In operation **1104**, a projector medallion may be formed of a translucent material may be attached to the necklace strand. In operation **1106**, a battery compartment may be connected to the backside interior within the interior. In operation **1108**, a battery serving as a power source may be placed within the battery compartment. In operation **1110**, a control chip may be electrically connected to the battery. In operation **1112**, the control chip may be secured to a chip mount within the interior. In operation **1114**, the projector light may be mounted to a projector light connector that connects the projector light to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the projector light.

(115) In operation **1116**, the wire may be bridged to the control chip via a wire connector. In operation **1118**, an optical conduit may be placed within the interior. In operation **1120**, a proximal support, a medial support, and a distal support may be attached to the backside interior of the projector medallion to support the optical conduit at an oblique angle to the backside interior. In operation **1122**, an emblem may be integrated onto the emblem lens. In operation **1124**, an opening may be cut into the projector medallion on at least one of the front side of the projector medallion and the underside of the projector medallion. In operation **1126**, A securing bracket may be attached to the optical conduit that at least partially secures the optical conduit within the interior by at least partially affixing the optical conduit to the projector light connector. In operation **1128**, a plurality of necklace LEDs may be attached to the wire. In operation **1130**, the necklace LEDs may be encased into the plurality of necklace bulbs, the necklace bulbs further comprising a bulb interior, the bulb interior further comprising an LED mount. In operation **1132**, a medallion light may be mounted to a medallion light connector that connects the medallion light to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the medallion light.

(116) Walking at night in the dark may be inherently dangerous, but it may be even more so on holiday nights. On holidays such as Halloween and New Years, drivers may be more likely to be under the influence and pedestrians may be more likely to be dressed in dark, hard-to-see clothing. Despite the considerable risks and dangers involved with walking in public on Halloween and New Years, children continue to dress in dark costumes and ignore the need for personal visual indicators. Children, while dressed in their Halloween attire, often fail to obey traffic and/or pedestrian laws and ordinances. Children may also often fail to use common sense such as looking both ways before crossing a road and/or using a crosswalk. These failures may result not only from a lack of knowledge of laws and the rules of the road, but also from the excitement that holiday

festivities may bring. Children may be playing with friends, searching for their next house to trick-or-treat at, and/or running around aimlessly as a result of their elevated blood sugar.
(117) As a result of failing to properly obey the rules of the road and/or use common walking sense, children may become the victim of traffic accidents. The embodiments of FIGS. **1**A-**10**B may remedy the risks involved with Halloween and New Years night and help to protect children from themselves and from the hidden dangers around them. The embodiments of FIGS. **1**A-**10**B produce lighting that may be seen by those driving automobiles, which may increase the likelihood the child is seen by the driver which may decrease the likelihood of an accident. The embodiments of FIGS. **1**A-**10**B may enable children and pedestrians to supplant the use of items such as hand held flash lights, headlamps, reflective vests, and/or reflective headwear. The embodiments of FIGS. **1**A-**10**B may enable users to have free hands while using the device, which may enable them to participate in other activities more easily.
(118) Although the present embodiments have been described with reference to specific example embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader spirit and scope of the various embodiments.
(119) A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the claimed invention. In addition, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other embodiments are within the scope of the following claims.
(120) It may be appreciated that the various systems, methods, and apparatus disclosed herein may be embodied in a machine-readable medium and/or a machine accessible medium compatible with a data processing system (e.g., a computer system), and/or may be performed in any order.
(121) The structures and modules in the figures may be shown as distinct and communicating with only a few specific structures and not others. The structures may be merged with each other, may perform overlapping functions, and may communicate with other structures not shown to be connected in the figures. Accordingly, the specification and/or drawings may be regarded in an illustrative rather than a restrictive sense.

Claims

1. An apparatus comprising: a wire serving as a necklace strand; a projector medallion formed of a translucent material comprising a backside, a front side, an underside, and an interior, wherein the projector medallion is connected to the necklace strand; a power source in the form of a battery within the interior of the projector medallion; a projector light within the interior of the projector medallion; a control chip within the interior of the projector medallion that controls power from the battery to the projector light; an optical conduit within the interior of the projector medallion comprising an emblem lens and a refraction means designed to manipulate the light emanating from the projector light, wherein the projector light is at least partially encased within the optical conduit to allow light to travel through a light canal within the optical conduit and pass through the emblem lens and the refraction means, wherein the refraction means is at an end of the optical conduit and is adjacent to the underside of the projector medallion, wherein the emblem lens is within the optical conduit and is positioned to receive light from the projector light, an emblem integrated onto the emblem lens; and an opening of the projector medallion cut into the projector medallion, wherein the refraction means is adjacent to the opening, wherein the optical conduit and the opening are substantially planar to one another, and wherein the opening allows light to project from the projector light to an area exterior to the refraction means without being obstructed by the translucent material of the projector medallion.

2. The apparatus of claim 1 further comprising: a medallion light within the interior of the projector medallion, wherein the medallion light emits light through the translucent material of the projector medallion to the area exterior to the projector medallion.
3. The apparatus of claim 1 further comprising: a wire connector designed for mechanically and electrically connecting the wire to the projector medallion via the control chip, thereby facilitating the transmission of power from the control chip to the wire while ensuring the wire's integration into the electrical system of the apparatus.
4. The apparatus of claim 1 further comprising: a projector light connector that provides both mechanical support and electrical connection for the projector light to the control chip, enabling power transfer from the control chip to the projector light and ensuring the projector light is securely positioned within the interior of the projector medallion.
5. The apparatus of claim 2 further comprising: a medallion light connector that provides both mechanical support and electrical connection for the medallion light to the control chip, enabling power transfer from the control chip to the medallion light and ensuring the medallion light is securely positioned within the interior of the projector medallion.
6. The apparatus of claim 4 further comprising: a securing bracket attached to the optical conduit that at least partially secures the optical conduit within the interior of the projector medallion by at least partially affixing the optical conduit to the projector light connector.
7. The apparatus of claim 1 further comprising: a proximal support, a medial support, and a distal support attached to a backside interior of the projector medallion to support the optical conduit at an oblique angle to the backside interior, wherein the proximal support is shorter than the medial support, wherein the medial support is shorter than the distal support, wherein the medial support comprises a medial cutout to receive and support the optical conduit, wherein the medial support is closer to the underside of the projector medallion than the proximal support, wherein the distal support is closer to the underside of the projector medallion than the medial support and the proximal support, and wherein the distal support comprises a distal cutout to receive and support the optical conduit.
8. The apparatus of claim 1 further comprising: a plurality of necklace LEDs attached to the wire, wherein the plurality of necklace LEDs receive power from the battery via the wire connector and the wire.
9. The apparatus of claim 1 wherein the optical conduit is polyhedral in shape.
10. The apparatus of claim 1 wherein the optical conduit is cylindrical in shape.
11. The apparatus of claim 1 wherein when the light travels through the light canal, the light passes through the emblem lens and is at least partially obstructed by the emblem thereby creating a decorative projection to the area exterior to the projector medallion after passing through the refraction means.
12. An apparatus comprising: a wire serving as a necklace strand; a projector medallion formed of a translucent material comprising a backside, a front side, an underside, and an interior, wherein the projector medallion is substantially connected to the wire; a battery compartment within the interior of the medallion; a battery serving as a power source within the battery compartment; a control chip within the interior of the projector medallion that is electrically connected to the battery and controls power to a plurality of projector lights, wherein the plurality of projector lights are positioned within the interior, a plurality of projector light connectors that connect the plurality of projector lights to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the plurality of projector lights, wherein the plurality of projector light connectors support the plurality of projector lights within the interior; a wire connector that connects the wire to the projector medallion via the control chip, wherein the wire connector allows for power to move from the battery to the control chip and then to the wire; a plurality of optical conduits within the interior, each of which comprising a light canal, an emblem lens, and a refraction means, wherein each individual optical conduit houses at least one of the

plurality projector lights, an emblem integrated onto each emblem lens, wherein when the light travels through the light canal of the plurality of optical conduits, the light passes through the emblem lens and is at least partially obstructed by the emblem creating a decorative projection at an area exterior to the projector medallion after passing through the refraction means; a plurality of openings of the projector medallion on at least one of the front side of the projector medallion and the underside of the projector medallion, wherein each individual refraction means is adjacent to at least one of the openings, wherein each individual optical conduit of the plurality of optical conduits is substantially planar to one of the plurality of openings, and wherein each of the plurality of openings allow light to project from the plurality of projector lights to the area exterior to the projector medallion without being obstructed by the translucent material of the projector medallion.

13. The apparatus of claim 12 further comprising: a medallion light that is positioned within the interior and emits light through the translucent material of the projector medallion to the area exterior to the projector medallion, wherein the medallion light is electrically connected to the battery.

14. The apparatus of claim 13 further comprising: a medallion light connector that connects the medallion light to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the medallion light, wherein the medallion light connector supports the medallion light within the interior.

15. The apparatus of claim 12 further comprising: a plurality of necklace LEDs that are attached to the wire, wherein the plurality of necklace LEDs are electrically connected to the battery via the necklace strand.

16. The apparatus of claim 12 further comprising: a securing bracket attached to the plurality of optical conduits that at least partially secures the plurality of optical conduits within the interior by at least partially affixing the plurality of optical conduits to the plurality of projector light connectors.

17. The apparatus of claim 12 further comprising: a plurality of proximal supports, a plurality of medial supports, and a plurality of distal supports attached to a backside interior of the projector medallion to support the plurality of optical conduits at an angle oblique to the backside interior, wherein the plurality of proximal supports are shorter than the plurality of medial supports and are closer to the battery compartment than the plurality of medial supports, wherein the plurality of medial supports are shorter than the plurality of distal supports and are closer to the battery compartment than the plurality of distal supports, wherein each of the plurality of medial supports comprise a medial cutout to receive and support at least one of the plurality of optical conduits, wherein the plurality of distal supports are closer to the underside of the projector medallion than the plurality of medial supports and the plurality of proximal supports, wherein the plurality of medial supports are closer to the underside of the projector medallion than the plurality of proximal supports, and Wherein each of the plurality of distal supports comprise a distal cutout to receive and support at least one of the plurality of optical conduits.

18. The apparatus of claim 12 further comprising: a plurality of necklace LEDs attached to the wire, wherein the necklace LEDs receive power from the battery via the control chip which then directs power to the wire connector which is electrically connected to the wire.

19. The apparatus of claim 12 further comprising: a plurality of necklace bulbs encasing the necklace LEDs, the necklace bulbs further comprising a bulb interior, the bulb interior further comprising an LED mount, wherein the plurality of necklace bulbs are formed from the translucent material, wherein one of the plurality of necklace LEDs is affixed within the bulb interior of one of the plurality of necklace bulbs via the LED mount, and wherein the plurality of necklace LEDs project light through the necklace bulbs to the area exterior to the bulb.

20. A method comprising: forming a necklace strand from a wire, wherein the wire is capable of transmitting an electrical current; attaching a projector medallion formed of a translucent material to the necklace strand, wherein the projector medallion comprises a backside, a front side, an

underside, and an interior, wherein the backside comprises a backside interior that faces the interior of the projector medallion; connecting a battery compartment to the backside interior within the interior; placing a battery serving as a power source within the battery compartment, wherein the battery is accessible through the backside; electrically connecting a control chip to the battery, wherein the control chip is within the interior of the projector medallion; securing the control chip to a chip mount within the interior, wherein the chip mount is attached to the backside interior and is adjacent to the battery compartment, wherein the control chip controls power from the battery to a projector light, and wherein the projector light is positioned within the interior of the medallion; mounting the projector light to a projector light connector that connects the projector light to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the projector light, wherein the projector light connector supports the projector light within the interior; bridging the wire to the control chip via a wire connector, wherein the wire connector mounts the wire to the projector medallion at the control chip to form the necklace strand, wherein the wire connector allows for power to move from the battery to the control chip and then to the wire; placing an optical conduit within the interior, wherein the optical conduit comprises a light canal, an emblem lens, and a refraction means, wherein the optical conduit is at least one of polyhedral in shape and cylindrical in shape, wherein the projector light is at least partially encased within the optical conduit, wherein light from the projector light travels through the light canal of the optical conduit and passes through the emblem lens and the refraction means to an area exterior to the refraction means, wherein the refraction means is at an end of the optical conduit and is adjacent to the underside of the projector medallion, and wherein the emblem lens is within the optical conduit at an area less close to the end of the optical conduit than the refraction means; attaching a proximal support, a medial support, and a distal support, to the backside interior of the projector medallion to support the optical conduit at an oblique angle to the backside interior, wherein the proximal support is shorter than the medial support, wherein the medial support is shorter than the distal support, wherein the medial support comprises a medial cutout to receive and support the optical conduit, wherein the medial support is closer to the underside of the projector medallion than the proximal support, wherein the distal support is closer to the underside of the projector medallion than the medial support and the proximal support, and wherein the distal support comprises a distal cutout to receive and support the optical conduit, integrating an emblem onto the emblem lens, wherein when the light travels through the light canal, the light passes through the emblem lens and is at least partially obstructed by the emblem creating a decorative projection at the area exterior to the projector medallion after passing through the refraction means; cutting an opening into the projector medallion on at least one of the front side of the projector medallion and the underside of the projector medallion, wherein the refraction means is adjacent to the opening, wherein the optical conduit and the opening are substantially planar to one another, wherein the opening of the projector medallion is a shape that substantially corresponds to the shape of the end of optical conduit and the refraction means, and wherein the opening allows light to project from the projector light to an area exterior to the projector medallion without being obstructed by the translucent material of the projector medallion.

21. The method of **20** further comprising: attaching a securing bracket to the optical conduit that at least partially secures the optical conduit within the interior by at least partially affixing the optical conduit to the projector light connector.

22. The method of **21** further comprising: attaching a plurality of necklace LEDs to the wire, wherein the necklace LEDs receive power from the battery via the wire connector and the wire.

23. The method of claim 22 further comprising: encasing the necklace LEDs into the plurality of necklace bulbs, the necklace bulbs further comprising a bulb interior, the bulb interior further comprising an LED mount, wherein the plurality of necklace bulbs are formed from the translucent material, wherein one of the plurality of necklace LEDs is affixed within the bulb interior of one of the plurality of necklace bulbs via the LED mount, and wherein the plurality of necklace LEDs

project light through the necklace bulbs to the area exterior to the bulb.
24. The method of claim 23 further comprising: mounting a medallion light to a medallion light connector that connects the medallion light to the control chip of the projector medallion and allows for power to move from the battery to the control chip and then to the medallion light, wherein the medallion light connector supports the medallion light within the interior.